

Revision Notes for Class 8 Science Chapter 7 – Particulate Nature of Matter

Matter is all around us, making up everything from grains of sand to water and even air. Everything in the universe is made up of extremely small particles called constituent particles. These particles are too small to see with our naked eyes, but their arrangement and behaviour decide whether a substance is a solid, liquid, or gas.

When we break chalk or dissolve sugar in water, we see that these materials can be split into very fine particles. On dissolving sugar, the sweetness spreads across the water, even though the sugar itself appears to have disappeared. This demonstrates that matter is made up of small, invisible particles that can mix and occupy spaces between other particles.

Constituent Particles of Matter

The smallest pieces of substances like chalk, sand, or sugar are called constituent particles. These can't be broken down further by physical means. Even though "particles" can sometimes mean dust or pollution, in science, particles mean the actual tiny building blocks—atoms and molecules.

These constituent particles are held together by certain forces called interparticle attractions. The nature and strength of these attractions determine the physical state—solid, liquid, or gas—of the substance.

States of Matter

There are mainly three states of matter: solid, liquid, and gas. The difference between these states lies in how tightly or loosely the particles are packed, how they move, and how strong the forces between them are.

Solid State

In solids, particles are packed very closely with very strong attractions between them. This tight packing ensures that solids have a fixed shape and volume. The particles do not move freely, but they do vibrate at their positions. Examples include iron nails, stones, and wood. When heat is applied, the particles vibrate even faster, and with enough energy, the solid melts into a liquid at a specific temperature called the melting point.

- Solids have fixed shape and volume.
- Particles are very closely packed and can only vibrate in place.
- The melting point is the temperature at which a solid turns to liquid. For example, ice melts at 0°C , urea at 133°C , and iron at 1538°C .

Liquid State

In liquids, particles are less tightly packed than in solids, and their attractions are weaker. Liquids can flow and take the shape of any container but maintain a fixed volume. Liquid particles move around each other, explaining why liquids like water or milk can flow and be poured. On heating, when these particles gain enough energy, they can change into the gaseous state at their boiling point. Some particles even escape at lower temperatures through evaporation.

- Liquids have fixed volume, but not fixed shape.
- Particles are more loosely packed than solids and can move freely.
- Liquids can flow and are called fluids.

Gaseous State

Gases do not have a fixed shape or fixed volume. The particles in gases are very far apart and move freely in all directions. The attractions between them are so weak that they can fill any space available. Because of this, gases can mix quickly and spread out

to fill a room. For example, the fragrance of an incense stick can be smelled all around a room because gas particles are always moving and dispersing.

- No fixed shape and no fixed volume.
- Particles are very far apart and move rapidly.
- Interparticle attractions are negligible.

Comparison: Interparticle Spacing and Attractions

The space between particles (interparticle spacing) is least in solids, more in liquids, and greatest in gases. This explains why solids are incompressible, liquids are almost incompressible, and gases are highly compressible. Activities such as using a syringe show that air can be compressed easily, while water resists compression.

Comparison of Solids, Liquids, and Gases

Property	Solid	Liquid	Gas
Interparticle spacing	Minimum	More than solids	Maximum
Packing of particles	Closely packed	Loosely packed	Free
Interparticle attraction	Maximum	Less than solids	Negligible
Movement of particles	Vibrate only	Move freely within liquid	Move freely everywhere

Movement and Mixing of Particles

Particles in all three states are always in motion, but the speed varies. In experiments, when potassium permanganate is placed in water, the color spreads, showing that particles move and mix on their own. The rate at which particles move increases with temperature, explaining why substances mix faster in warm water compared to cold.

In gases, the movement is fastest. For instance, the smell of incense quickly fills a room because gas particles travel and distribute rapidly. This property of gases and liquids to move and mix is called diffusion.

Important Points to Remember

- Matter is made up of tiny particles called atoms and molecules.
- Solid, liquid, and gas states differ in the arrangement and behaviour of these particles.
- Particles are held together by forces that are strongest in solids, weaker in liquids, and weakest in gases.
- Heating increases particle movement and can cause change from solid to liquid (melting) or liquid to gas (boiling/evaporation).
- Both liquids and gases flow and are called fluids.
- Gases can be compressed easily due to large spaces between particles; solids and liquids cannot.

Quick Questions for Revision

- Why can water flow but a wooden block cannot?
- What happens at the melting and boiling points of a substance?
- How do the arrangements of particles differ in solids, liquids, and gases?
- Why does sugar seem to disappear on dissolving, but makes the water sweet?

Some Activities You Can Try

- Press air inside a syringe and notice it compresses. Try the same with water and observe the difference.
- Observe how the smell of camphor or incense spreads faster when the windows are open (more air movement).

- Try dissolving salt and sand separately in water and notice which one disappears and why.

Snapshots

- Matter is made up of very small particles that are in constant motion.
- **Solids:** closely packed, strong attractions, definite shape and volume.
- **Liquids:** less tightly packed, weaker attractions, fixed volume but no fixed shape.
- **Gases:** particles far apart, move fast, negligible attractions, no fixed shape or volume.

Interesting Fact

Long ago, Indian philosopher Acharya Kanad spoke about the "Parmanu" or atom— the tiniest unit of matter. A substance's properties come from how these particles are grouped and how they move.